

Operational and Preventive Maintenance Checklist

General Outline for Plant Equipment Inspection Frequency

Operational and Preventive Maintenance	Frequency						
	Daily	Weekly	Monthly	3Mon	6Mon	Yearly	As Necessary
1 PLANT AREA/TANKFARM/BOND WALL							
a. Check factory wall, roof, floor	X						
b. Check floors at equipment support legs	X						
c. Check plant area	X						
d. Check for unusual odor/smell	X						
e. Check to ensure good housekeeping	X						
f. Check bond wall for leakage	X						
g. Check any evidence of bond wall erosion	X						
h. Lookout for any damage on bond wall	X						
i. Check for water grown weeds	X						
k. Fence damage	X						
m. Check any evidence of short circuiting	X						
2 PIPING/INLET PIPING							
a. Check bolting, flanges and gaskets for leaks	X						
b. Check piping system support for any damage	X						
c. Check piping surface for wears/damage/leaks	X						
d. Check screens, strainers, and filter							X
e. Check/verify inlet flowmeter for accuracy	X						
3 CENTRIFUGAL PUMPS							
a. Check the overall sound level of the pump	X						
b. Check the suction filter and clean							X
c. Check temperature of pump bearing	X						
d. Lubricate pump							X
e. Check bearing vibration	X						
f. Check oil level	X						
g. Identify any leakage	X						
h. Check alignment of pump with motor shaft	X						
i. Check the operating pressure	X						
j. Check the heating efficiency of the pump	X						
4. ELECTRIC MOTORS							
a. Check bearing for abnormal temp. & vibration	X						
b. Check oil level of bearing		X					
c. identify any leakage	X						
d. Check for abnormal sound	X						
e. Check for unusual noise	X						
5. PREHEATER/REACTOR/DISTILLATION/DESULPHURIZATION TOWER							
a. Check sludge discharge valve for blockage		X					
b. Check sight glass for any damage or cracks	X						
c. Check sight glass for reduced visibility			X				
d. Check all field mounted instrument and gauges		X					
e. Check heater wires for any short-circuiting	X						
f. Check discharge temperature gauge	X						
g. Check for unprotected electrical connections	X						
h. Check for unusual noise or vibrations	X						
i. Check gas leak around welded or flanged joints	X						
j. Check sludge piping for leak or damage	X						
k. Check sludge temp. gauge for accuracy		X					

6. VALVES a. Check the torque of the handwheel or lever b. Check lubrication of lever bearing or spindle c. Check full handle stroke or turn SAFETY/RELIEF VALVES a. Check general condition of valve b. Verify consistency of required valve setting c. Check indications on valve name plate							
	X						
		X					
	X						
	X						
	X						
	X						
7. HEAT EXCHANGER/COOLING WATER a. Check for any damage on the insulated portion b. Check for leak around the shell wall c. Check associated equipment of proper operation d. Check cooling water color for unusual change e. Check pressure and temperature differential f. Check inlet flow for correct readings							
	X						
	X						
	X						
	X						
	X						
	X						
8. GAS/LIQUID SEPARATOR a. Check sight glass for visibility in reading b. Check pressure gauge is within desired pressure c. Check for leakage or escape of liquid or gas d. Check for any foaming on sight glass							
	X						
	X						
	X						
	X						
9. ELECTRICAL CONTROL PANEL a. Check electrical timers for correct readings b. Observe controllers and settings for malfunction c. Check for unprotected electrical connections d. Check panel housing for damage or short circuit							
	X						
	X						
	X						
	X						
10. PLANT STORAGE VESSELS a. Check sight glass for cracks or damage b. Check shell walls/top/bottom for leaks/damage c. Check support legs for corrosion/wear/damage d. Check all associated gauges for cleanliness							
	X						
	X						
	X						
	X						

The maintenance department will also make an inspection on the receipt of a report of a potentially hazardous condition made by the plant operating personnel.

Appropriate records will be kept to identify the inspection made, deficiencies found, and action taken to correct such deficiencies.